

EE2080: Random Processes

Lecture 1: Introduction

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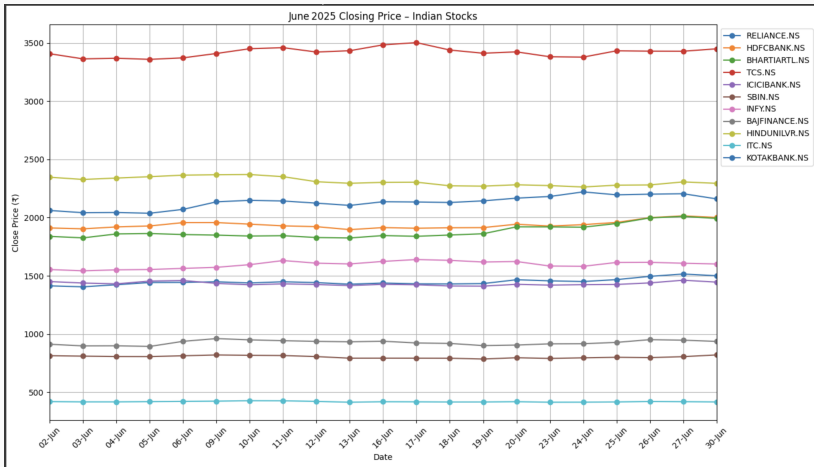
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Random Processes: Definition

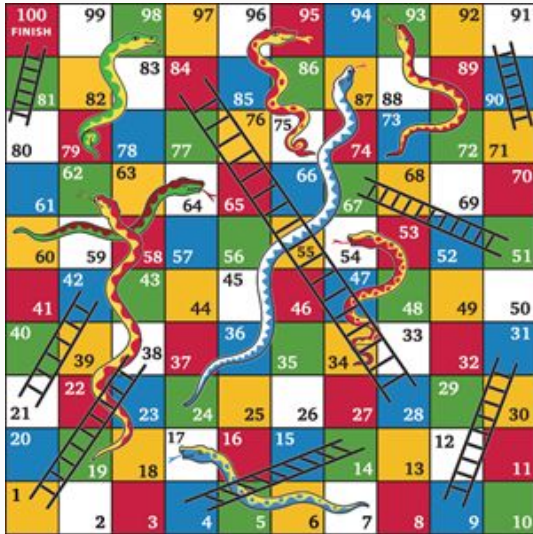
A random process is an indexed collection $X : \{X_t\}_{t \in T}$ of random variables all defined w.r.t the same probability space.

- The index set T is often interpreted as time.
- If T is finite, X is a random vector
- If T is countably infinite: Discrete Random Process
- If T is uncountably infinite: Continuous Random Process

Example 1: Stock Market Fluctuations



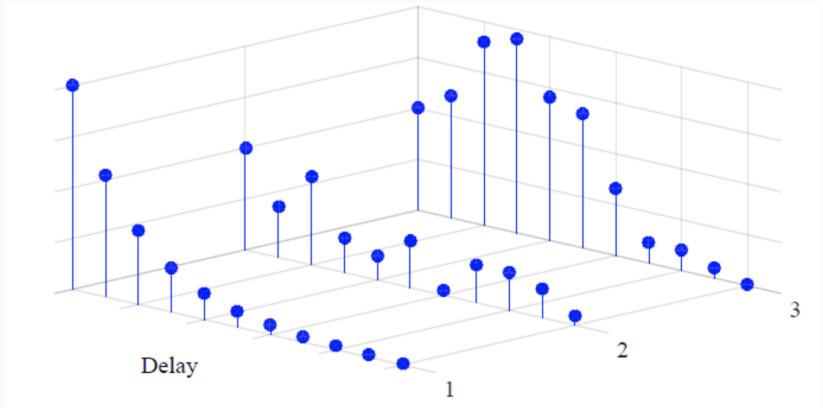
Example 2: Snake and Ladder Game



Example 3: PageRank

PageRank Explained: A Stochastic Approach

Example 4: Channel Impulse Response of a Time-Varying Multi-path Wireless Channel



Course Logistics

1. **Random processes:** Introduction, Examples, Finite Dimensional Distributions, Properties: Mean, Autocorrelation, Autocovariance, Stationarity, WSS Process, Ergodicity, Stationary and Independent Increments.
2. IID Bernoulli process, Poisson Process, Gaussian Processes, Random Walks, filtrations, stopping time, Wald's Lemma
3. **Discrete time Markov chains (DTMCs):** state space, examples, transition probability matrix (TPM), time homogeneous Markov chains, Chapman–Kolmogorov equation for TH-DTMCs, Strong Markov property, hitting and recurrence times.
4. Linear Systems Input/Output Relationships for WSS processes, Power Spectral Density, White Noise (Spectrally flat) processes Representation of random signals - Karhunen-Loeve Expansion.

- Programming Assignment (Python): 20 marks
- Mid-segment Exam: 30 marks
- End-segment Exam: 50 marks